JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

JNTUK R23 Regulation · Semester I

SUBJECT: BEEE – BASIC ELECTRICAL & ELECTRONICS ENGINEERING

Basic Electronic Circuits & Instrumentation

UNIT – V

Rectifiers · Amplifiers · Instrumentation

◆ DC Power Supply & Block Diagram

◆ Full Wave Bridge Rectifier & Capacitor Filter

◆ Zener Voltage Regulator

◆ Public Address System & CE Amplifier

◆ Frequency Response · Electronic Instrumentation System

Exam-Oriented Notes · All Diagrams Included · Complete Unit Coverage

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01 Introduction to Electronic Circuits

Electronics is the branch of physics and engineering that deals with the flow and control of electrons in vacuum, gases, or semiconductors. **Electronic circuits** use components such as diodes, transistors, capacitors, and resistors to process and control electrical signals for various applications such as communication, computation, and instrumentation.

1.1 CLASSIFICATION OF ELECTRONIC CIRCUITS

Type	Function	Examples
Power Circuits	Convert AC to DC power	Rectifiers, Regulators
Amplifier Circuits	Increase signal strength	CE Amplifier, PA System
Oscillator Circuits	Generate periodic waveforms	RC, LC Oscillators
Logic Circuits	Perform digital operations	Gates, Flip-flops
Instrumentation Circuits	Measure physical quantities	Sensors, Signal conditioners

1.2 IMPORTANCE OF ELECTRONIC CIRCUITS

- Electronic circuits form the backbone of all modern electrical appliances and systems.
- They enable conversion of AC mains supply into regulated DC required by gadgets.
- Amplifiers make weak signals from microphones or sensors usable for further processing.
- Instrumentation systems help in precise measurement and monitoring of physical parameters.
- Without electronic circuits, modern communication, computing, and control systems would not exist.

1.3 KEY TERMS

Term	Meaning
Diode	P-N junction device that allows current in one direction only
Transistor	Three-terminal semiconductor device used for amplification and switching
Rectification	Process of converting AC into DC
Regulation	Maintaining constant output voltage despite variations in load/input

Gain	Ratio of output signal to input signal amplitude
Bandwidth	Range of frequencies over which an amplifier operates effectively

📌 **Note:** Unit V covers three important domains: Power Supply circuits (Rectifiers & Regulators), Amplifiers (CE Amplifier, PA System), and Electronic Instrumentation systems — all of which are core to BEEE syllabus and frequently examined.

02 Power Supply Overview

2.1 DEFINITION

A **power supply** is an electronic circuit that converts the AC mains voltage (230V, 50Hz) into a smooth and regulated DC voltage suitable for powering electronic circuits and devices.

2.2 NEED FOR DC POWER SUPPLY

Most electronic circuits require a stable DC supply voltage. The AC mains supply cannot be directly used for transistors, ICs, or sensors. Therefore, we need a power supply that:

- Steps down the high voltage AC (230V) to required lower voltage levels.
- Converts AC to pulsating DC using rectifier diodes.
- Smooths the pulsating DC into almost constant DC using filters.
- Maintains constant output voltage using voltage regulators.

2.3 STAGES OF DC POWER SUPPLY

AC Mains 230V, 50Hz → Transformer (Step Down) → Rectifier (AC→DC) → Filter (Smooth DC) → Regulator (Stable DC) → Load (Output)

2.4 TYPES OF RECTIFIERS

Type	Diodes Used	Output	Efficiency
Half Wave	1	Half cycles only	~40.6%
Full Wave (Centre Tap)	2	Both cycles	~81.2%
Full Wave Bridge	4	Both cycles	~81.2%

KEY COMPARISON FORMULAS

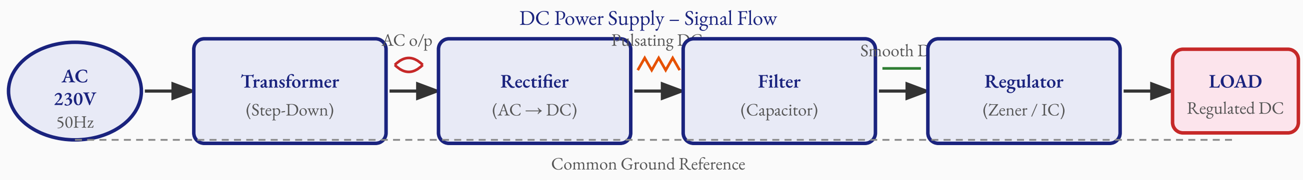
```
Ripple Factor (HWR) = 1.21
Ripple Factor (FWR) = 0.48
```

```
Efficiency (FWR) = 81.2%
```

03 DC Power Supply – Block Diagram

3.1 BLOCK DIAGRAM WITH EXPLANATION

FIG. 3.1 – BLOCK DIAGRAM OF DC POWER SUPPLY



The DC power supply converts 230V AC into regulated DC through four stages: Transformer → Rectifier → Filter → Regulator.

3.2 DESCRIPTION OF EACH BLOCK

a) Transformer (Step-Down)

The step-down transformer reduces the 230V AC mains to a lower AC voltage (e.g., 12V or 9V AC) suitable for rectification. It provides electrical isolation between mains and the circuit, ensuring safety.

b) Rectifier

The rectifier converts the AC voltage from the transformer into pulsating DC voltage. In a bridge rectifier, all four diodes conduct alternately during both half-cycles, giving a full-wave output.

c) Filter

A capacitor filter smooths the pulsating DC by charging during peaks and discharging during valleys, reducing ripple and producing a nearly steady DC output. The larger the capacitor, the smoother the output.

d) Regulator

The voltage regulator maintains a constant output voltage despite variations in load current or input voltage. A Zener diode regulator is a simple and widely used shunt regulator that clamps the output to the Zener breakdown voltage.

OUTPUT AT EACH STAGE

```
Transformer O/P : V_s = V_p × (N_s / N_p)
```

```
Rectifier O/P : V_dc = 0.636 × V_m (FWR)
Regulated O/P : V_out = V_Z (Zener voltage)
```

04 Full Wave Bridge Rectifier

4.1 DEFINITION

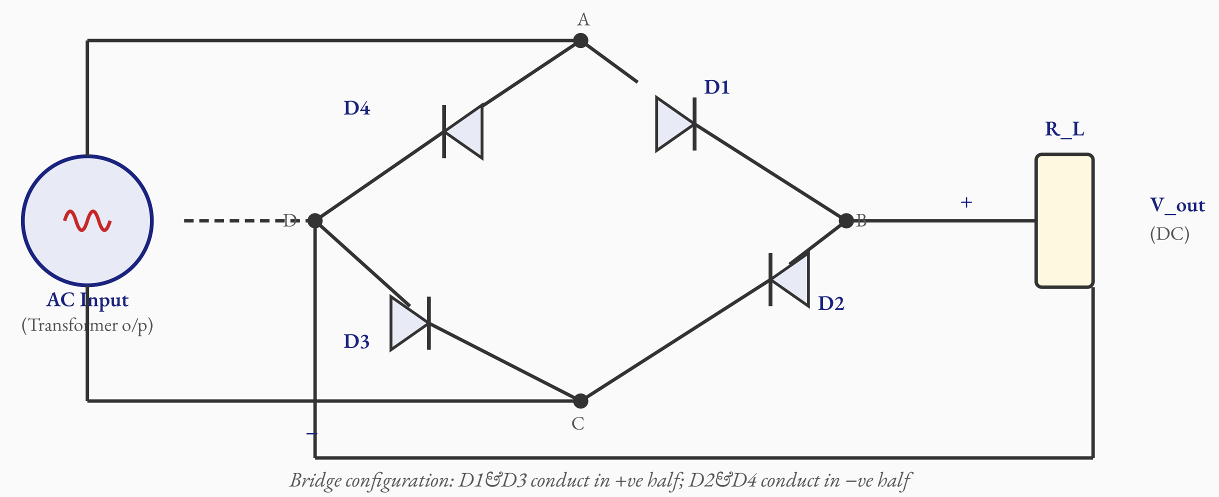
A **Full Wave Bridge Rectifier** is a circuit that converts both positive and negative half-cycles of AC into pulsating DC, using four diodes arranged in a bridge configuration. It is the most widely used rectifier circuit.

4.2 CONSTRUCTION

The bridge rectifier consists of:

- **Four Diodes (D1, D2, D3, D4)** – connected in a bridge (diamond) configuration.
- **Transformer** – provides AC input voltage.
- **Load Resistor (R_L)** – connected across the output terminals.
- No centre-tap transformer is required (unlike the 2-diode FWR).

FIG. 4.1 – FULL WAVE BRIDGE RECTIFIER CIRCUIT



Bridge Rectifier with 4 diodes D1–D4, AC input, and load resistor R_L.

4.3 WORKING PRINCIPLE

During Positive Half-Cycle (A is +ve, C is -ve):

- Diodes **D1 and D3** are forward biased and conduct.
- Current flows: A → D1 → B → R_L → D → D3 → C.
- Current through R_L is from top to bottom, producing positive output.

- Diodes D2 and D4 are reverse biased and do not conduct.

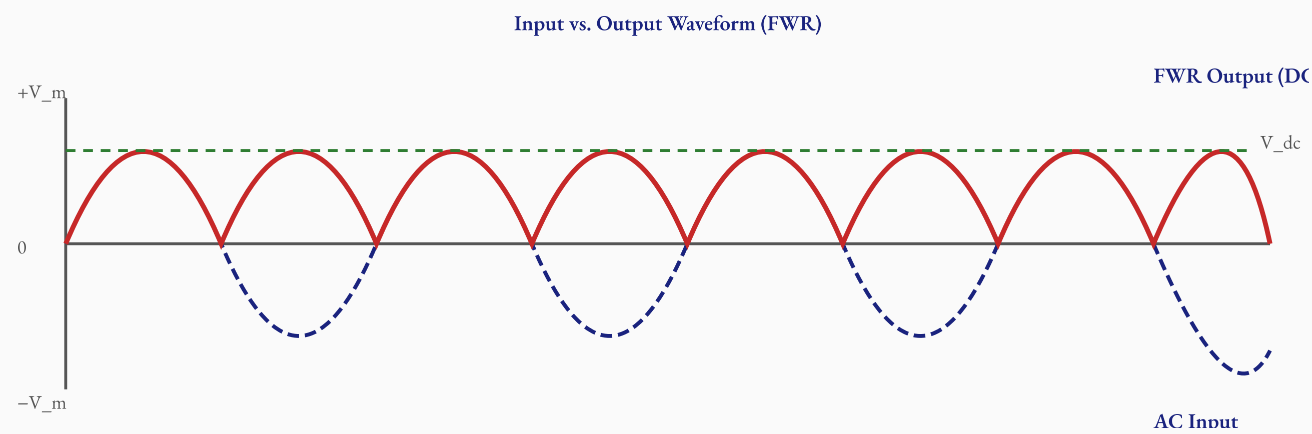
During Negative Half-Cycle (C is +ve, A is −ve):

- Diodes **D2 and D4** are forward biased and conduct.
- Current flows: C → D2 → B → R_L → D → D4 → A.
- Current through R_L is again from top to bottom — same direction.
- D1 and D3 are reverse biased and do not conduct.

> **Key Observation:** In both half-cycles, current through the load R_L flows in the same direction. This is the fundamental principle of full-wave rectification.

4.4 OUTPUT WAVEFORM

FIG. 4.2 – INPUT AC AND OUTPUT DC WAVEFORMS



Blue dashed: AC input. Red solid: Full-wave rectified output. Green dashed: Average DC value.

4.5 KEY FORMULAS

FULL WAVE BRIDGE RECTIFIER – FORMULAS

```
V_dc = (2 × V_m) / π ≈ 0.636 V_m
V_rms = V_m / √2 ≈ 0.707 V_m
Ripple Factor (γ) = 0.48
Efficiency (η) = 81.2%
PIV (per diode) = V_m
```

4.6 ADVANTAGES, DISADVANTAGES & APPLICATIONS

Advantages	Disadvantages

• Higher efficiency (81.2%) than HWR • No centre-tap transformer needed • Lower ripple factor (0.48) • Better output utilization	• Uses 4 diodes (more cost & complexity) • Voltage drop across 2 diodes at any time • Not suitable for very low voltage outputs

Applications: Battery chargers, DC power supplies, SMPS circuits, welding equipment, and portable electronic devices.

05 Capacitor Filter

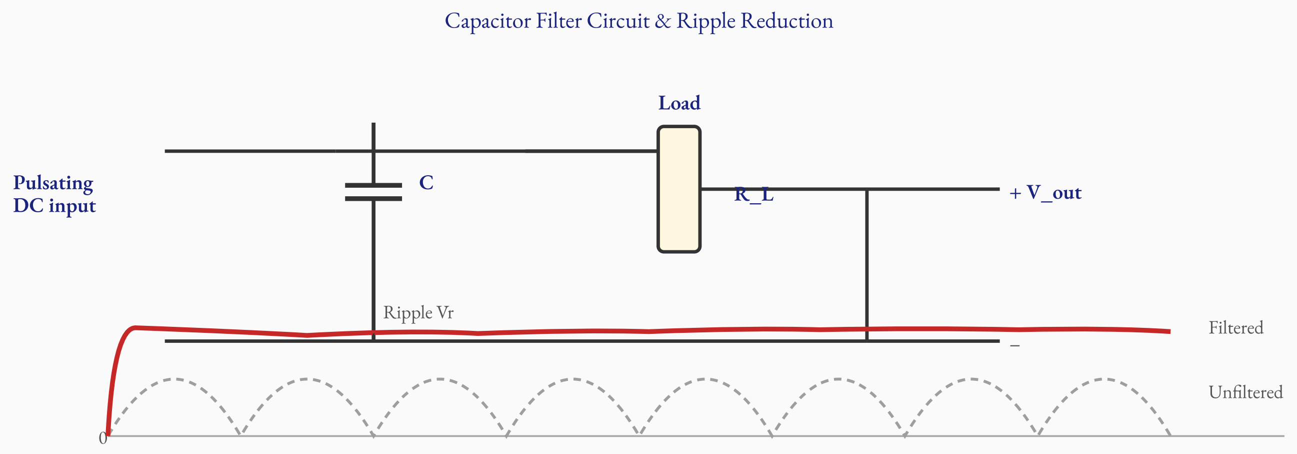
5.1 DEFINITION

A **capacitor filter** is a circuit element (capacitor) connected in parallel with the load resistor across the output of the rectifier. It reduces the ripple in the pulsating DC and provides a smoother, nearly constant DC output.

5.2 WORKING PRINCIPLE

- **Charging Phase:** When the rectified output voltage rises above the capacitor's stored voltage, the capacitor charges rapidly through the diodes to the peak voltage V_m.
- **Discharging Phase:** When the rectified voltage falls below the capacitor voltage, the diodes become reverse biased and stop conducting. The capacitor slowly discharges through the load, maintaining the output voltage.
- This charge-discharge cycle repeats at twice the supply frequency (100 Hz for 50 Hz input), resulting in very small ripple.
- A **larger capacitance** means slower discharge and hence less ripple.

FIG. 5.1 – CAPACITOR FILTER CIRCUIT AND OUTPUT WAVEFORM



The capacitor charges to peak and discharges through load, maintaining near-constant DC. Red = filtered output; Grey dashed = unfiltered pulsating DC.

5.3 KEY FORMULAS

CAPACITOR FILTER FORMULAS

```
Ripple Factor (γ) = 1 / (4√3 × f × C × R_L)
Ripple Voltage (V_r) = I_dc / (4 × f × C)
V_dc ≈ V_m - V_r/2
```

```
where f = supply frequency, C = capacitance, R_L = load
resistance
```

5.4 ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Simple, low cost; Large reduction in ripple; Easy to add to any rectifier circuit	Output voltage drops under heavy load; Capacitor causes high peak currents; Not suitable for precision regulation

Applications: Power supply filters in radios, TVs, audio equipment, mobile chargers, and laboratory DC supplies.

06 Zener Voltage Regulator

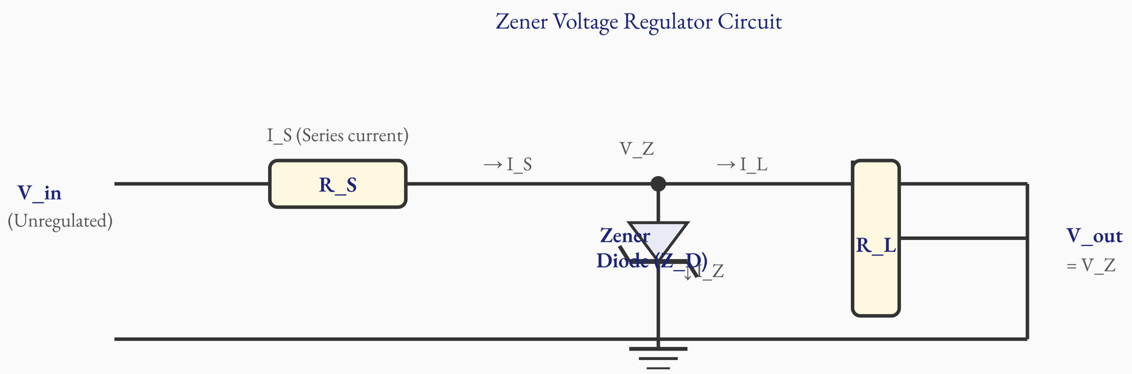
6.1 DEFINITION

A **Zener voltage regulator** is a shunt-type voltage regulator circuit that uses a Zener diode operating in its reverse breakdown region to maintain a constant output voltage across the load, even when the input voltage or load current varies.

6.2 CONSTRUCTION

- **Zener Diode (Z_D):** A specially doped P-N junction diode designed to operate in the reverse breakdown region at a specific voltage called the Zener voltage (V_Z).
- **Series Resistor (R_S):** Connected in series between the input and output; limits the current and drops the excess voltage.
- **Load Resistor (R_L):** Connected in parallel with the Zener diode at the output.
- **Input Supply (V_in):** Unregulated DC from the filter stage.

FIG. 6.1 – ZENER VOLTAGE REGULATOR CIRCUIT



Zener diode connected in shunt (parallel with load). Series resistor R_S drops excess voltage. Output = V_Z (constant).

6.3 WORKING PRINCIPLE

- The unregulated DC (V_in) is applied to the circuit. The Zener diode is reverse biased.
- When V_in exceeds the Zener breakdown voltage V_Z, the Zener diode conducts in reverse, clamping the output voltage to V_Z.
- **If load R_L decreases (more current drawn):** The current through Zener (I_Z) decreases, but V_Z remains constant. Total current I_S = I_Z + I_L remains roughly constant.

- **If V_in increases:** The excess voltage is dropped across R_S, and I_Z increases, but V_out = V_Z stays constant.
- Thus, the Zener diode acts as a voltage clamp, maintaining V_out = V_Z despite variations in V_in or I_L.

6.4 KEY FORMULAS

ZENER REGULATOR FORMULAS

```
V_out = V_Z (Zener breakdown voltage)
I_S = (V_in − V_Z) / R_S
I_L = V_Z / R_L
I_Z = I_S − I_L
Power dissipated by Zener: P_Z = V_Z × I_Z
```

6.5 ADVANTAGES, DISADVANTAGES & APPLICATIONS

Advantages	Disadvantages
Simple and inexpensive; No complex feedback circuit; Good for low-power applications; Stable output voltage	Low efficiency due to power dissipation in R_S; Not suitable for high load currents; Output changes slightly with temperature

Applications: Reference voltage generation, low-power DC regulated supplies, overvoltage protection circuits, voltage-level detection, and battery-operated devices.

07 Amplifier Basics

7.1 DEFINITION

An **amplifier** is an electronic circuit that increases the amplitude (voltage, current, or power) of an input signal without significantly altering its waveform shape. The transistor is the key active element used in most amplifier circuits.

7.2 CLASSIFICATION OF AMPLIFIERS

Basis of Classification	Types
Based on signal type	Voltage amplifier, Current amplifier, Power amplifier
Based on coupling	RC coupled, Transformer coupled, Direct coupled
Based on configuration	Common Emitter (CE), Common Base (CB), Common Collector (CC)
Based on class	Class A, Class B, Class AB, Class C

7.3 KEY PARAMETERS OF AMPLIFIER

AMPLIFIER PARAMETERS

```
Voltage Gain (A_v) = V_out / V_in
Current Gain (A_i) = I_out / I_in
Power Gain (A_p) = P_out / P_in = A_v × A_i
Gain in dB = 20 × log10(A_v) [voltage gain]
Bandwidth (BW) = f_H − f_L
```

7.4 COMMON EMITTER (CE) CONFIGURATION – OVERVIEW

In the CE configuration, the **emitter terminal is common** to both input and output circuits. The input signal is applied between base and emitter; the output is taken between collector and emitter. The CE amplifier is the most widely used configuration because it provides both high voltage gain and high current gain, and has moderate input and output impedances.

Parameter	CE Amplifier	CB Amplifier	CC Amplifier
Voltage Gain	High (>1)	High (>1)	<1

Current Gain	High (β)	<1	High (β+1)
Phase shift	180°	0°	0°
Input Impedance	Medium	Very Low	Very High
Application	General amplifier	RF amplifier	Buffer/emitter follower

🔴 **Important for Exam:** CE amplifier introduces a 180° phase shift between input and output. The output is an inverted version of the input signal.

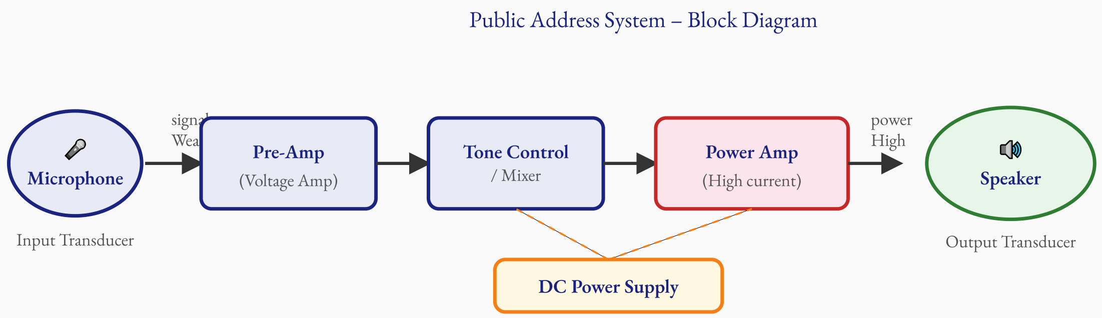
08 Public Address System

8.1 DEFINITION

A **Public Address (PA) System** is an electronic audio system used to amplify sound from a source (microphone, audio player) and reproduce it at a sufficiently high power level through loudspeakers so that a large audience can hear it clearly.

8.2 BLOCK DIAGRAM OF PA SYSTEM

FIG. 8.1 – BLOCK DIAGRAM OF PUBLIC ADDRESS SYSTEM



Sound → Microphone → Pre-Amp → Tone Control/Mixer → Power Amplifier → Speaker → Sound (amplified)

8.3 DESCRIPTION OF EACH BLOCK

a) Microphone (Input Transducer)

Converts sound waves (acoustic energy) into corresponding weak electrical signals (typically a few millivolts). This is the input transducer of the PA system.

b) Pre-Amplifier (Voltage Amplifier)

Amplifies the weak signal from the microphone to a sufficient voltage level. It is a small-signal amplifier using BJT or FET. It adds no significant distortion and has high input impedance.

c) Tone Control / Mixer

Allows adjustment of the frequency response (bass, treble, midrange) of the audio signal. In a mixer, multiple input sources (e.g., music, voice) can be combined before amplification.

d) Power Amplifier

Amplifies the voltage signal to sufficient power levels (watts) required to drive the loudspeaker. It has low output impedance matched to the speaker's impedance (typically 4 Ω or 8 Ω). Class A, B, or AB configurations are used.

e) Speaker (Output Transducer)

Converts the amplified electrical signal back into sound waves (acoustic energy) that can be heard by the audience. It is an electromagnetic transducer with a voice coil and cone.

f) DC Power Supply

Provides the necessary regulated DC voltages to power all the amplifier stages. It converts the AC mains to appropriate DC levels.

8.4 APPLICATIONS OF PA SYSTEMS

- Public speeches, conferences, and auditoriums.
- Schools, colleges, and sports stadiums.
- Music concerts and entertainment venues.
- Train stations, airports, and public announcement systems.
- Religious places and community halls.

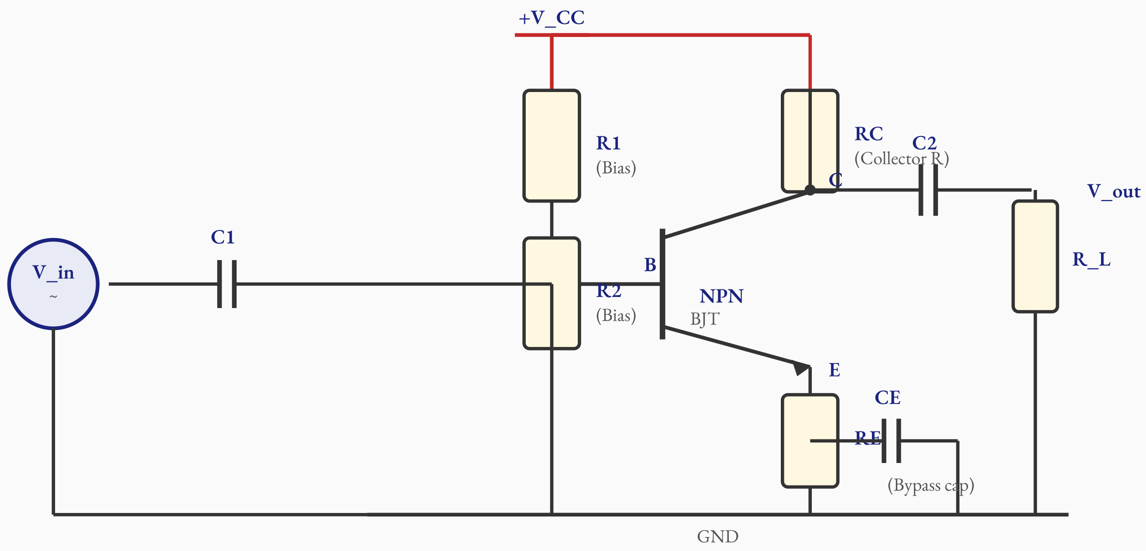
09 RC Coupled Common Emitter Amplifier

9.1 DEFINITION

An **RC Coupled Common Emitter (CE) Amplifier** is a small-signal voltage amplifier circuit in which a transistor operates in the common emitter configuration, and coupling capacitors (C1, C2) are used to connect the input signal and output load while blocking DC components.

9.2 CONSTRUCTION & CIRCUIT DIAGRAM

FIG. 9.1 – RC COUPLED COMMON EMITTER AMPLIFIER CIRCUIT



RC Coupled Common Emitter Amplifier

NPN transistor in CE configuration with R1, R2 bias divider, RC collector resistor, RE emitter resistor, CE bypass capacitor, C1 input coupling, and C2 output coupling capacitors.

9.3 COMPONENT FUNCTIONS

Component	Function
R1, R2 (Voltage Divider)	Establish stable DC bias (Q-point) for the transistor
RC (Collector Resistor)	Convert collector current variations into voltage signal
RE (Emitter Resistor)	Provide DC stability and temperature stabilization
C1 (Input Coupling Cap)	Block DC from input source; allow only AC signal to base

C2 (Output Coupling Cap)	Block DC from collector; allow only AC signal to load
CE (Bypass Capacitor)	Bypass RE for AC signal, increasing AC gain
V_CC	DC supply voltage for biasing and amplification

9.4 WORKING PRINCIPLE

- The transistor is biased in the active region through R1, R2 voltage divider so that it amplifies AC signals.
- The input AC signal (V_in) is coupled through C1 to the base of the transistor.
- Small variations in base current cause large variations in collector current ($I_C = \beta \times I_B$).
- The collector current variations cause voltage variations across RC, producing an amplified but **180° phase-inverted** output.
- The output signal is coupled through C2 to the load R_L.
- CE bypasses RE for AC signals, so RE does not reduce AC gain.

CE AMPLIFIER GAIN FORMULAS

```
Voltage Gain: A_v = -β × RC / r_e ≈ -RC / R_E (approx)
Input Impedance: Z_in = R1 ‖ R2 ‖ β × r_e
Output Impedance: Z_out ≈ RC
r_e = 26mV / I_C (AC emitter resistance)
Phase shift = 180° (output inverted w.r.t. input)
```

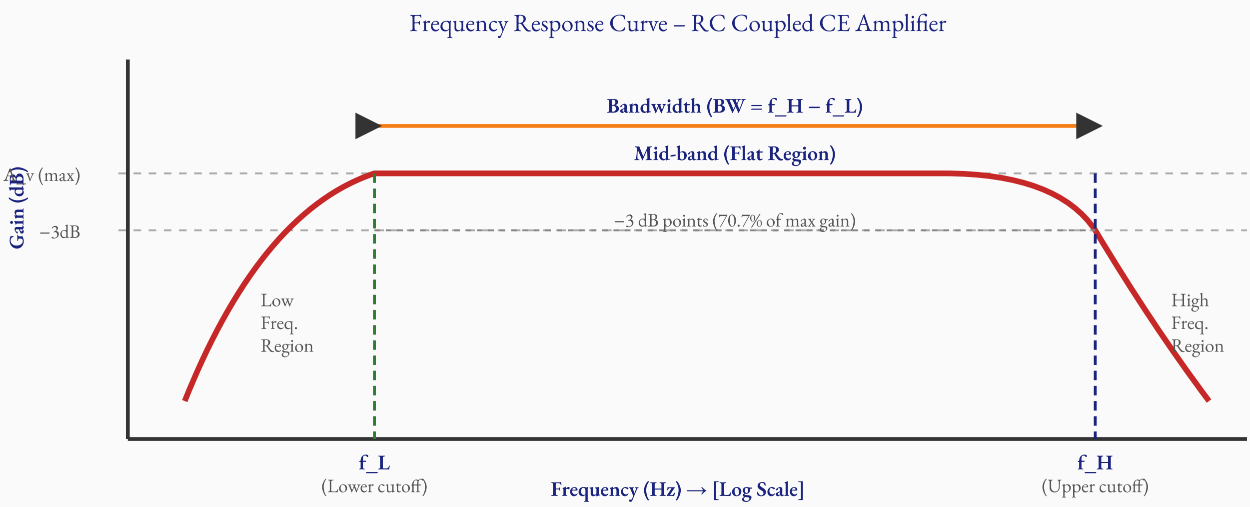
10 Frequency Response of CE Amplifier

10.1 DEFINITION

The **frequency response** of an amplifier is a graph showing how the voltage gain (in dB) varies with the frequency of the input signal. It indicates the useful bandwidth of the amplifier — the range of frequencies over which the amplifier maintains its gain within acceptable limits.

10.2 FREQUENCY RESPONSE CURVE

FIG. 10.1 – FREQUENCY RESPONSE CURVE OF RC COUPLED CE AMPLIFIER



Mid-band region has constant gain. Gain drops at low and high frequencies. f_L and f_H are half-power (−3dB) points. BW = f_H − f_L.

10.3 THREE FREQUENCY REGIONS

a) Low Frequency Region ($f < f_L$)

At low frequencies, the reactance of coupling capacitors (C1, C2) is very high ($X_C = 1/2\pi fC$). This means the capacitors block part of the signal, reducing the gain. The gain falls off as frequency decreases. The lower cut-off frequency f_L is where the gain drops to 70.7% of midband gain (i.e., −3 dB point).

b) Mid-band Region ($f_L < f < f_H$)

In this region, coupling capacitors act as short circuits (very low reactance) and transistor junction capacitances are negligible. The gain is maximum and nearly constant. This is the useful operating range of the amplifier. Voltage gain $A_v \approx -R_C/r_e$.

c) High Frequency Region ($f > f_H$)

At high frequencies, the internal capacitances of the transistor (C_{be} and C_{bc}) become significant and start to short-circuit the signal path. These capacitances shunt the signal to ground, causing gain to fall. The upper cut-off frequency f_H is where gain again drops to −3 dB.

FREQUENCY RESPONSE FORMULAS

```
Bandwidth (BW) = f_H - f_L
Gain at cut-off = A_v / √2 = 0.707 × A_v_mid
Gain in dB at cut-off = A_v_mid(dB) - 3 dB
f_L ≈ 1 / (2π × R × C_coupling)
f_H ≈ 1 / (2π × R × C_transistor)
```

📌 **Exam Tip:** The bandwidth of an amplifier determines its fidelity. Audio amplifiers need BW of 20 Hz to 20 kHz to reproduce sound faithfully. Always label f_L, f_H, −3dB points and BW on the frequency response graph.

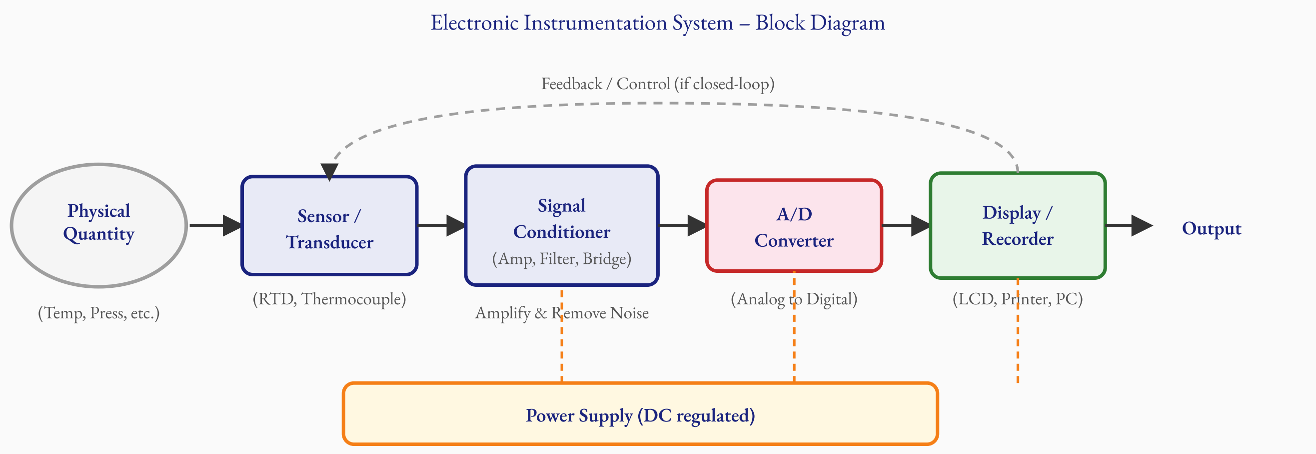
11 Electronic Instrumentation System

11.1 DEFINITION

An **electronic instrumentation system** is a complete measurement chain that senses a physical quantity (temperature, pressure, flow, etc.), converts it into an electrical signal, processes and conditions that signal, and finally displays or records the measured quantity in a readable form.

11.2 BLOCK DIAGRAM

FIG. 11.1 – BLOCK DIAGRAM OF ELECTRONIC INSTRUMENTATION SYSTEM



Physical Qty → Sensor/Transducer → Signal Conditioner → A/D Converter → Display/Recorder. Dashed: feedback loop in closed-loop systems.

11.3 DESCRIPTION OF EACH BLOCK

a) Physical Quantity (Measurand)

The physical quantity to be measured — such as temperature, pressure, flow, speed, displacement, or humidity. This is the input to the instrumentation system.

b) Sensor / Transducer

Converts the physical quantity into a proportional electrical signal (voltage or current). Examples: RTD and thermocouple for temperature; strain gauge for pressure; photodiode for light; accelerometer for vibration. The output is usually a very small electrical signal (millivolts).

c) Signal Conditioner

Processes the weak sensor signal to make it suitable for display or conversion. It performs: **amplification** (increase signal level), **filtering** (remove noise), **bridge circuits** (for resistive sensors), **linearization** (correct non-linear sensor response), and **isolation** (galvanic isolation for safety).

d) Analog-to-Digital (A/D) Converter

Converts the conditioned analog signal to a digital value that can be processed by a microcontroller or computer. The resolution (number of bits) determines measurement accuracy.

e) Display / Recorder / Output

Presents the measured value in a human-readable form. Types: LCD digital display, analog meter, chart recorder, printer, or computer interface (USB, RS232). In control systems, the output may also be used to generate a control signal.

11.4 EXAMPLES OF INSTRUMENTATION SYSTEMS

Measured Quantity	Sensor Used	Signal Conditioner	Display
Temperature	Thermocouple / RTD	Instrumentation Amp	Digital LCD
Pressure	Strain Gauge	Wheatstone Bridge + Amp	Pressure gauge / LCD
Light Intensity	LDR / Photodiode	Op-amp circuit	Lux meter
Speed (RPM)	Tachometer	Frequency counter	Digital RPM meter

12 Comparison Tables

12.1 COMPARISON OF RECTIFIERS

Parameter	Half Wave Rectifier	Full Wave (Centre Tap)	Full Wave Bridge
No. of Diodes	1	2	4
Transformer	Simple	Centre-tap required	Simple (no centre-tap)
V_{dc}	$0.318\ V_m$	$0.636\ V_m$	$0.636\ V_m$
Ripple Factor	1.21	0.48	0.48
Efficiency	40.6%	81.2%	81.2%
PIV per diode	V_m	$2\ V_m$	V_m
Output frequency	f (supply freq)	2f	2f
Cost	Lowest	Medium	Slightly higher (4 diodes)
Application	Simple, low-power	General power supply	Most common power supply

12.2 COMPARISON OF TRANSISTOR AMPLIFIER CONFIGURATIONS

Parameter	Common Emitter (CE)	Common Base (CB)	Common Collector (CC)
Input Terminal	Base	Emitter	Base
Output Terminal	Collector	Collector	Emitter
Common Terminal	Emitter	Base	Collector
Voltage Gain	High (>1)	High (>1)	<1 (≈1)
Current Gain	High (β)	<1 (α)	High (β+1)
Power Gain	Highest	Medium	Medium
Input Impedance	Medium (1–5 kΩ)	Very Low (30–150 Ω)	Very High (100 kΩ+)

Output Impedance	Medium–High	High	Very Low (25–50 Ω)
Phase Shift	180°	0°	0°
Use	General amplifier	RF/high-freq amplifier	Buffer, impedance matching

12.3 COMPARISON OF FILTER TYPES

Parameter	Capacitor Filter (Shunt C)	Inductor Filter (Series L)	π-Filter (LC)
Circuit Complexity	Simple	Simple	Complex
Cost	Low	Medium	High
Ripple Reduction	Good	Moderate	Excellent
Suitable Load	Light load (high R_L)	Heavy load (low R_L)	Any load
Common Use	Power supplies for ICs	Audio amplifiers	Laboratory power supplies

13 Questions & Answers

PART A – 2 MARK QUESTIONS (SHORT ANSWER)

Q1. Define ripple factor. 2M

Ripple Factor (γ) is defined as the ratio of the RMS value of the AC component (ripple voltage, V_r) present in the rectified output to the average (DC) value of the output voltage (V_dc). It is a measure of the purity of the DC output.

```
γ = V_r(rms) / V_dc
```

A lower ripple factor means a purer (better) DC output. For a half-wave rectifier, $\gamma = 1.21$; for a full-wave rectifier, $\gamma = 0.48$, indicating FWR produces a better quality DC output than HWR.

Q2. What is meant by Zener breakdown voltage? 2M

The **Zener breakdown voltage (V_Z)** is the specific reverse bias voltage at which a Zener diode undergoes controlled avalanche/Zener breakdown and starts conducting a large reverse current while maintaining a nearly constant voltage across its terminals.

Below V_Z, only a tiny leakage current flows. Once V_Z is reached, the diode conducts heavily in reverse, and the voltage across it remains clamped at V_Z regardless of the current. This property makes the Zener diode ideal for voltage regulation. V_Z is a design parameter and is available from 2.4V to 200V in standard Zener diodes.

Q3. What is the bandwidth of an amplifier? 2M

The **bandwidth (BW)** of an amplifier is defined as the range of frequencies over which the amplifier can amplify signals while maintaining its gain within 3 dB (70.7%) of its maximum midband gain.

```
BW = f_H − f_L
```

where f_L is the lower cut-off frequency and f_H is the upper cut-off frequency — both are −3 dB points. For an audio amplifier, BW ≈ 20 Hz to 20 kHz. A larger bandwidth means the amplifier can amplify a wider range of signal frequencies faithfully.

Q4. State the function of a coupling capacitor in a CE amplifier. 2M

A **coupling capacitor** (C1 at input, C2 at output) in a CE amplifier serves two main functions:

1. **DC Blocking:** It blocks the DC bias voltages of one stage from affecting the adjacent stage. This ensures the correct Q-point (operating point) of the transistor is maintained without interference from the source or load.
2. **AC Signal Coupling:** It allows the AC signal to pass from the source to the transistor input (C1) and from the transistor output to the load (C2), since the capacitor offers low reactance at signal frequencies.

Thus, coupling capacitors decouple the DC operating points while coupling the AC signals between stages.

Q5. What is a transducer? Give an example. 2M

A **transducer** is a device that converts one form of energy into another, specifically a physical quantity (non-electrical) into a proportional electrical signal (or vice versa). In instrumentation, transducers are used as the primary sensing elements that convert measurands into electrical signals for processing.

Examples:

- Thermocouple – converts temperature difference into millivoltage.
- Microphone – converts sound pressure into electrical voltage.
- Strain gauge – converts mechanical strain into change in resistance.
- Piezoelectric crystal – converts mechanical pressure into voltage.

PART B – 5 MARK QUESTIONS (DESCRIPTIVE)

Q6. Explain the working of a Full Wave Bridge Rectifier with a neat circuit diagram. 5M

Definition: A Full Wave Bridge Rectifier converts both positive and negative half-cycles of AC into pulsating DC using four diodes in a bridge configuration.

Circuit: Four diodes D1, D2, D3, D4 are connected in a bridge. AC supply from the transformer is applied to two opposite nodes of the bridge. Output is taken from the other two nodes across load R_L.

Working during Positive half-cycle (A positive, C negative):

- D1 (A→B) and D3 (D→C) are forward biased.
- Current path: A → D1 → B → R_L → D → D3 → C (back to transformer).
- Current flows top-to-bottom through R_L.

- D2 and D4 are reverse biased — no conduction.

Working during Negative half-cycle (C positive, A negative):

- D2 (B→C) and D4 (D→A) are forward biased.
- Current path: C → D2 → B → R_L → D → D4 → A.
- Current again flows top-to-bottom through R_L — same direction!
- D1 and D3 are reverse biased.

Result: Both half-cycles contribute to unidirectional current through R_L, giving full-wave rectified output.

Key Parameters:

- V_dc = 0.636 V_m | Ripple Factor = 0.48 | Efficiency = 81.2%
- PIV per diode = V_m | Output frequency = 2f

Advantages over HWR: Double the output frequency makes filtering easier; higher efficiency; no centre-tap transformer required.

Q7. Explain the Zener Voltage Regulator with circuit diagram and working. 5M

Zener Voltage Regulator: Uses a Zener diode in reverse breakdown to maintain constant output voltage.

Circuit Description: Series resistor R_S connects input V_in to the output node. Zener diode Z_D is connected in shunt (parallel) between the output node and ground (reverse biased). Load R_L is also in parallel with Z_D.

Working Principle:

- When V_in ≥ V_Z, the Zener diode breaks down and maintains V_out = V_Z.
- If V_in increases: extra voltage drops across R_S; I_Z increases; V_out stays at V_Z.
- If load R_L decreases (more current drawn): I_L increases; I_Z decreases; but V_out = V_Z is maintained since I_S = I_Z + I_L is kept constant by R_S.

Current equations:

- I_S = (V_in – V_Z) / R_S (series current)
- I_L = V_Z / R_L (load current)
- I_Z = I_S – I_L (Zener current)

Condition: I_Z must remain between I_Z(min) and I_Z(max) for proper regulation. R_S must be chosen accordingly.

Advantages: Simple, inexpensive, no complex feedback. **Disadvantage:** Power loss in R_S; efficiency is low; not suitable for heavy loads.

Applications: Reference voltage generation, overvoltage protection, portable device power supplies, and voltage clamping circuits.

Q8. Describe the RC Coupled Common Emitter Amplifier and its frequency response. 5M

RC Coupled CE Amplifier: The most popular small-signal amplifier using an NPN (or PNP) transistor in common emitter configuration. Coupling capacitors C1 and C2 connect the input signal and output load while blocking DC.

Components and Functions:

- *R1, R2:* Voltage divider bias — sets stable DC operating point (Q-point).
- *RC:* Collector resistor — converts current changes to voltage output.
- *RE:* Emitter resistor — improves temperature stability.
- *CE:* Bypass capacitor — short-circuits RE for AC to maximize AC gain.
- *C1, C2:* Coupling capacitors — pass AC, block DC.

Working: Input signal applied to base through C1 → small variation in I_B → large variation in I_C ($= \beta \times I_B$) → voltage drop across RC changes → amplified output taken across RC via C2. Output is 180° phase inverted.

Frequency Response:

- *Low Frequency:* Coupling capacitor reactance is high → gain falls below f_L.
- *Mid-band:* Coupling caps act as shorts → gain is maximum and flat. $A_v = RC/r_e$.
- *High Frequency:* Transistor internal capacitances shunt the signal → gain falls above f_H.

Key Parameters: $BW = f_H - f_L$; Gain at −3 dB points = $0.707 \times A_{v\_mid}$; Phase shift = 180°.

Q9. Draw the block diagram of a DC power supply and explain each stage. 5M

A DC power supply converts 230V AC mains into a regulated DC voltage. It has four main stages:

1. Transformer (Step-Down): Reduces 230V AC to a lower value (e.g., 12V AC) using electromagnetic induction. The turns ratio N_s/N_p determines the step-down factor. Also provides electrical isolation.

2. Rectifier: Converts AC to pulsating DC. Bridge rectifier uses 4 diodes — D1, D2 conduct during positive half-cycle; D3, D4 during negative half-cycle. Both half-cycles contribute to output. $V_{dc} = 0.636\, V_m$.

3. Filter (Capacitor Filter): A large capacitor is connected in parallel with the load. It charges to peak during diode conduction and discharges slowly through load between peaks, smoothing the pulsating DC. The larger the capacitor, the smaller the ripple.

4. Regulator (Zener Regulator): A Zener diode clamps the output to V_Z despite variations in load or input. Series resistor R_S drops the excess voltage. Output $V_{out} = V_Z$ = constant.

Signal flow: 230V AC → Transformer → Low AC → Rectifier → Pulsating DC → Filter → Smooth DC → Regulator → Regulated DC → Load.

Q10. Explain the Electronic Instrumentation System with a block diagram. 5M

An electronic instrumentation system is a measurement chain that senses a physical quantity and converts it into a readable output.

Blocks and their functions:

1. *Physical Quantity (Measurand):* The input to be measured — temperature, pressure, flow, displacement, etc.
2. *Sensor/Transducer:* Converts physical quantity to electrical signal. E.g., Thermocouple → mV output for temperature; Strain gauge → resistance change for pressure. Output is usually weak (mV range).
3. *Signal Conditioner:* Amplifies the weak sensor signal; filters noise; linearises non-linear sensors; provides bridge excitation for resistive sensors. Output is a clean, strong analog signal (0–5V or 4–20mA).
4. *Analog-to-Digital (A/D) Converter:* Converts the analog signal to a digital number (binary). Resolution depends on number of bits (8-bit, 12-bit, 16-bit). Required for processing by digital systems.
5. *Display/Recorder:* Shows the measurement in human-readable form — LCD display, digital panel meter, chart recorder, or computer. In closed-loop systems, also generates a control signal.
6. *Power Supply:* Provides regulated DC power to all blocks of the system.

Examples: Digital thermometer (Thermocouple → Amp → ADC → LCD); Electronic blood pressure monitor; Industrial pressure gauge; Digital weighing scale.

14 Quick Revision Sheet – Unit V

🔌 DC POWER SUPPLY

- 4 Stages: Transformer → Rectifier → Filter → Regulator
- Transformer: steps down AC voltage
- Rectifier: AC → DC conversion
- Filter (Cap): smooths pulsating DC
- Regulator: maintains constant V_{out}

⚡ BRIDGE RECTIFIER

- 4 diodes in diamond configuration
- +ve cycle: D1, D3 conduct
- −ve cycle: D2, D4 conduct
- $V_{dc} = 0.636\, V_m$
- Ripple factor = 0.48
- Efficiency = 81.2%
- PIV = V_m (per diode)
- No centre-tap transformer needed

🔋 CAPACITOR FILTER

- C in parallel with R_L
- Charges during peak; discharges during valley
- Larger C → Less ripple
- $\gamma = 1 / (4\sqrt{3} \times f \times C \times R_L)$
- Output freq = 2f for FWR

🛡 ZENER REGULATOR

- Zener in reverse bias (shunt)
- $V_{out} = V_Z$ = constant
- $I_S = (V_{in} - V_Z) / R_S$
- $I_L = V_Z / R_L$
- $I_Z = I_S - I_L$
- $P_Z = V_Z \times I_Z$

📢 PA SYSTEM

- Mic → Pre-Amp → Tone Control → Power Amp → Speaker
- Mic: acoustic → electrical
- Speaker: electrical → acoustic
- Pre-Amp: voltage amplification
- Power Amp: current/power amplification

🔬 CE AMPLIFIER

- Phase shift = 180°
- $A_v = -RC / r_e$
- $r_e = 26mV / I_C$
- C1: input coupling (AC pass, DC block)
- C2: output coupling
- CE: bypass RE for AC gain
- R1, R2: DC bias divider

📊 FREQUENCY RESPONSE

- $BW = f_H - f_L$
- −3 dB → gain = $0.707 \times A_{v\_max}$

📡 INSTRUMENTATION SYSTEM

- Blocks: Sensor → Sig. Cond. → ADC → Display
- Sensor: physical qty → electrical

- Low freq: $C_{coupling}$ limits gain
- Mid-band: maximum flat gain
- High freq: transistor C_{BE}, C_{BC} limit gain
- Audio BW: 20 Hz – 20 kHz

- Signal Conditioner: Amp + Filter
- ADC: analog → digital
- Display: LCD / Recorder
- Ex: Thermocouple, Strain gauge, LDR

ALL IMPORTANT FORMULAS AT A GLANCE

Formula	Expression	Topic
V_{dc} (FWR)	$0.636 \times V_m$	Bridge Rectifier
V_{rms} (FWR)	$0.707 \times V_m$	Bridge Rectifier
Ripple Factor (FWR)	0.48	Bridge Rectifier
Efficiency (FWR)	81.2%	Bridge Rectifier
Ripple Factor (HWR)	1.21	Half Wave Rectifier
Ripple Factor (with filter)	$1/(4\sqrt{3} \times f \times C \times R_L)$	Capacitor Filter
Zener series current	$I_S = (V_{in} - V_Z)/R_S$	Zener Regulator
Load current	$I_L = V_Z/R_L$	Zener Regulator
Zener current	$I_Z = I_S - I_L$	Zener Regulator
Voltage Gain (CE)	$A_v = -RC/r_e$	CE Amplifier
r_e (AC emitter resistance)	$26mV / I_C$	CE Amplifier
Bandwidth	$BW = f_H - f_L$	Frequency Response
Gain at −3dB	$A_v / \sqrt{2} = 0.707 \times A_v$	Frequency Response
Gain in dB	$20 \times \log_{10}(A_v)$	Amplifier
Power Gain	$A_p = A_v \times A_i$	Amplifier

KEY POINTS TO REMEMBER FOR EXAM

🔴 **EXAM CRITICAL POINTS:**

- Bridge rectifier uses 4 diodes; no centre-tap transformer; PIV = V_m per diode.

- Ripple factor of FWR (0.48) is much better than HWR (1.21).
- Zener diode is always reverse biased in regulator circuits.
- CE amplifier gives 180° phase inversion (output is inverted).
- CE bypass capacitor CE increases AC gain by short-circuiting RE for AC signals.
- Frequency response curve: draw and label f_L, f_H, BW, −3dB points, midband.
- Bandwidth = f_H − f_L; −3dB point = 70.7% of maximum gain.
- Instrumentation system: Sensor → Signal Conditioner → ADC → Display.
- PA system: Microphone (input transducer) → Amplifiers → Speaker (output transducer).
- Capacitor filter: larger capacitance = smaller ripple = better filtering.

BEEE – Unit V Complete Notes

JNTUK R23 Regulation · Semester I · Subject Code: BEEE

All diagrams are original SVG drawings. Content prepared for exam-oriented study.